Venkata Lakshmi Parimala Pasupuleti

Senior Backend Engineer · AI Observability · Evaluation Platforms · Tracing · LangGraph · LangChain · Python · PostgreSQL · High-Throughput Systems · 9yr Production

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PROFESSIONAL SUMMARY

Senior backend engineer with 9+ years of production Python and a specialization in AI observability, evaluation platforms, and tracing systems - the exact backend LangSmith is built on. AgentProbe is the direct proof: I designed and shipped a production AI evaluation and observability platform end-to-end - 15-endpoint FastAPI backend, structured tracing across all agent steps, 5-metric evaluation engine (correctness, groundedness, hallucination rate, latency, token efficiency), LLM-as-Judge scoring, pass@k multi-trial harness, offline batch evaluation pipelines, and a CI-integrated regression gate blocking quality drift before deployment. I build with LangGraph and LangChain daily - as a power user of LangChain's own frameworks, I understand the tracing and evaluation pain that LangSmith solves from the inside. At Entrata I owned mission-critical Python backend systems for 7 years at 99.5% uptime - RabbitMQ async processing with retry logic and idempotency guarantees, 200+ PostgreSQL/MySQL query optimizations (30% performance gain), and AWS-native infrastructure serving 100K+ housing units. I know what high-throughput, reliable data services look like in production and what breaks them. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Backend & APIs: Python (9yr), FastAPI, Flask, REST APIs, API Design, Async Processing, RabbitMQ, Microservices, Service-Based Architecture, PostgreSQL, MySQL, Oracle, Redis, High-Throughput Systems, Performance Optimization, Query Optimization, Indexing, Partitioning, Stored Procedures, Distributed Systems, Reliability Engineering, Idempotency, Retry Logic, Dead-Letter Handling

AI Observability & Evaluation: LangChain, LangGraph, MCP, Observability, Tracing, Structured Logging, Evaluation Frameworks, LLM-as-Judge, Hallucination Detection, Groundedness, pass@k Harness, CI Regression Gate, Offline Batch Evaluation, Online Scoring, Drift Detection, Feedback Loops, Monitoring, Alerting, Agent Observability, AI Reliability, LLMOps, Anthropic API, OpenAI API, Ollama, ChromaDB, Vector Databases, RAG Pipelines

Cloud & Infrastructure: AWS (S3, EC2, RDS, Glue, Lambda, DynamoDB), Docker, GitHub Actions CI/CD, Automated Testing, Incident Response, Root-Cause Analysis, Runbooks, React.js, TypeScript, Linux, Bash, Git, Agile/Scrum, Technical Mentoring, Design Docs, Code Reviews

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer - Production Agentic AI Delivery

- Designed and deployed a production agentic AI backend (LangGraph, FastAPI, Python) as sole technical owner - architecture through implementation, evaluation pipeline, tracing infrastructure, and production deployment; built HITL guardrails and CI validation gates; delivered observability and feedback loop infrastructure for ongoing operation.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built ML pipelines (PyTorch, scikit-learn, R^2=0.977); designed observability dashboards (40% comprehension improvement); mentored 600+ students in Python, SQL, and distributed systems; cross-functional delivery across 3 research pipelines.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **High-throughput backend systems at production scale** - owned mission-critical Python backend systems for 7 years at 99.5% uptime serving 100K+ housing units; RabbitMQ event-driven async processing with retry logic, idempotency guarantees, and dead-letter handling; 200+ PostgreSQL/MySQL/Oracle query optimizations through execution plan analysis, indexing, and partitioning (30% performance gain); AWS S3/EC2/RDS/Glue infrastructure; Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); GitHub Actions CI/CD with automated testing.
- **Reliability, observability, and incident response** - built meaningful monitoring and alerting; authored runbooks; led incident response with RCA; built internal debugging tools reducing triage time 40%; kept bug count below 10 across 20+ production modules for 6+ consecutive months over 120+ Agile sprints. Fast feedback loops between production issues and code fixes - zero compliance violations across the entire ownership period.
- **Technical leadership** - led projects spanning multiple engineers; wrote design docs and architecture decisions; mentored 5+ engineers on system design and production practices; 300+ peer code reviews; Python/Bash automation eliminating 80% of manual workload.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (Python, PostgreSQL) for production SaaS; data pipeline for 1.6M records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Production AI Observability & Evaluation Platform

Observability · Evaluation · Tracing · CI Gate · LangSmith-equivalent · Open Source

Python · FastAPI · LangGraph · PostgreSQL · ChromaDB · Redis · GitHub Actions · Anthropic API · OpenAI API | github.com/venkatalakshmiparimala/AgentProbe

Production AI observability and evaluation backend - the same platform LangSmith provides to teams at scale. **Backend services:** 15-endpoint FastAPI backend with structured data services for evaluation runs, trace storage, metric aggregation, and provider routing; high-volume evaluation data storage and query optimization. **Observability & tracing:** structured tracing across all agent steps with full span capture; monitoring and alerting on eval quality metrics; feedback loops from production back into evaluation suites. **Evaluation engine:** 5-metric quality engine (correctness, groundedness, hallucination rate, latency, token efficiency); LLM-as-Judge scoring; pass@k multi-trial harness surfacing variance in non-deterministic outputs; offline batch evaluation pipelines. **Reliability:** CI-integrated regression gate (GitHub Actions) blocks quality drift before deployment - zero regressions across 1,240+ test cases. Open-source, multi-team deployable.

Leasel: Multi-Agent Production Backend - What LangSmith Traces at Scale

LangGraph · Multi-Agent · RAG · High-Throughput · Production

Python · LangGraph (8-agent) · FastAPI · PostgreSQL · SQLite · Redis · MCP · Anthropic API | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production multi-agent backend demonstrating exactly the system LangSmith must trace and evaluate at scale. **Backend architecture:** FastAPI backend powering 8-agent LangGraph StateGraph (Triage → Retrieval → Configuration → Historical → Structured Data → Reasoning → Validation → Escalation) with conditional routing; PostgreSQL + SQLite for shared agent state and knowledge persistence across sessions. **Performance & reliability:** confidence-based routing avoids unnecessary LLM calls; deterministic HITL guardrails enforce safe execution paths; structured output validation at every agent boundary. **Tracing:** every agent step emits structured trace data - span boundaries, confidence scores, source citations, and escalation signals - the production observability pattern AgentProbe tracks. 93.6% precision, 87.9% F1 across 1,240 test cases.

AI on FHIR: Full-Stack Backend - High-Volume API & Cloud Infrastructure

Backend · AWS · REST APIs · PostgreSQL · HIPAA · Performance

Python · Flask · React.js · TypeScript · Docker · AWS Lambda · PostgreSQL · REST APIs | github.com/venkatalakshmiparimala/AI_on_FHIR

Full-stack backend system on AWS for regulated healthcare data - HIPAA-compliant RBAC with full data isolation; REST API serving LLM-powered natural-language queries as structured FHIR API calls; AWS Lambda serverless with API gateway; PostgreSQL data services; complete audit trail and monitoring. Demonstrates backend reliability engineering, API design, and production observability in a high-stakes regulated environment; ~60% reduction in manual retrieval time.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017